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Abstract

FAIRE: Force-Aware Imitation with Reactive Execution for Robotic Polishing

Robotic polishing requires more than reaching a target region or completing a geometric path. Consistent surface finishing depends on reproducing continuous contact behavior, including the applied force trajectory, sliding velocity, and alignment between the polishing tool and the workpiece surface. A vision-only imitation policy can reproduce demonstrated motion, but it does not directly observe contact. Conversely, a force-aware policy must do more than recognize contact: it must convert the demonstrated interaction into a reference that can be executed safely and compliantly on a robot.

This thesis presents FAIRE, a precision-oriented framework for force-aware imitation with reactive execution in robotic polishing. FAIRE begins with a handheld teaching device equipped with a six-axis force/torque sensor and tracked in a virtual-reality coordinate system. An explicit VR-to-robot calibration transfers synchronized position, orientation, wrench, and image demonstrations into the robot workspace. A gated recurrent unit encodes a window of wrench measurements to represent contact evolution. Fused visual, proprioceptive, and force-history features are then used to predict short-horizon position and orientation references together with a desired force trajectory.

The learned desired force is a controller-level reference rather than a direct actuator command. During execution, an admittance controller converts the error between measured and desired force into a compliant Cartesian position correction. Two bounded residual reinforcement-learning policies operate

as precision-correction layers above this nominal execution: one scales the sliding velocity according to the contact state, and the other corrects the tool orientation to improve alignment between the tool-center-point normal and the estimated surface normal. The imitation policy and admittance controller remain responsible for task-level execution.

The evaluation is organized around three questions: whether a time-varying motion–force action improves contact reproduction, which force-history representation best tracks demonstrated force trajectories, and whether residual velocity and orientation corrections improve polishing precision. The protocol measures normal-force error, force-profile correlation, overshoot, contact loss, normal-alignment error, surface-quality improvement, spatial consistency, and task time on curved headlight surfaces. Because the source FAIRE manuscript does not yet contain completed measurements, quantitative conclusions and unverified implementation parameters are explicitly marked as planned or TBD rather than inferred.

Keywords: Robotic Polishing, Imitation Learning, Force/Torque Sensing, Force Control, Admittance Control, Residual Reinforcement Learning

Chapter 1

Introduction

1.1 Background and Motivation

Robotic polishing is a contact-rich manufacturing process in which the robot must coordinate geometric motion with the physical interaction between a compliant tool and a workpiece. A path that visits the intended surface region is necessary, but it is not sufficient for a consistent finish. The applied normal force, the sliding velocity, and the tool orientation jointly affect contact stability, dwell, and material removal. Errors in any of these variables can cause intermittent contact, excessive loading, nonuniform processing, or surface damage [1–3].

Classical polishing systems address this problem with geometric path planning and force or compliance control. Such systems can be highly repeatable when the workpiece geometry, contact model, fixture, and process parameters are known. Their performance nevertheless depends on accurate modeling and tuning. Local defects, fixture tolerances, uncertain compliance, and operator-specific adjustments are difficult to encode in a fixed analytic trajectory. These limitations motivate learning task-specific contact behavior from human demonstrations.

Imitation learning has become a practical method for acquiring visuomotor skills from demonstrations [4, 5]. In polishing, however, visual and kinematic observations alone do not uniquely describe the interaction. Similar tool poses can correspond to approach, stable contact, excessive loading, or contact loss. Force/torque sensing supplies direct evidence of this latent contact state.

Recent force-aware methods demonstrate the value of predicting or reacting to force during contact-rich manipulation [6–9]. Their existence also makes the scope of the present thesis precise: the contribution is not force sensing by itself, but its integration into a polishing-oriented demonstration, prediction, and compliant-execution framework.

1.2 From Motion Reproduction to Contact Reproduction

A polishing demonstration contains at least two coupled forms of information. The motion component describes where and how the tool should travel. The interaction component describes the force developed while that motion is executed. Replaying only the demonstrated path assumes that the robot and demonstrator encounter identical geometry and compliance. This assumption is fragile because a small surface-height error can substantially change the contact force without visibly changing the tangential path.

FAIRE therefore treats the desired force trajectory as part of the learned action. The force trajectory is not interpreted as a direct actuator-level command. Instead, a low-level admittance controller converts force-tracking error into a compliant Cartesian correction. This separation assigns complementary responsibilities to learning and control: imitation learning predicts the demonstrated task reference, while admittance control realizes that reference under local contact variation.

Contact is also temporal. An instantaneous wrench cannot always distinguish contact establishment from steady sliding or contact loss. FAIRE uses a gated recurrent unit (GRU) to encode a recent six-axis wrench history. The history representation is fused with visual context and robot state to predict short-horizon pose and force references. During execution, bounded

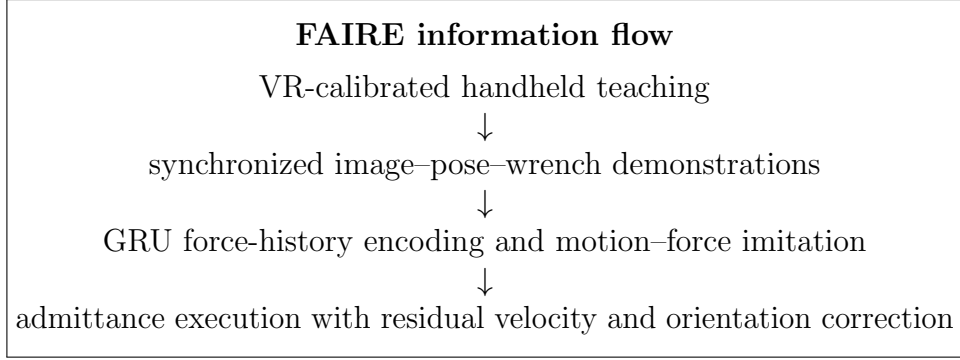


Figure 1.1: Conceptual organization of FAIRE. The final system illustration is planned; this box records only the verified information flow in the source manuscript.

residual reinforcement-learning (RL) policies adjust sliding velocity and tool orientation as precision corrections without replacing the nominal imitation policy.

1.3 Research Questions

The thesis is organized around three research questions inherited from the FAIRE manuscript.

RQ1: Does predicting a time-varying motion-force trajectory and realizing it through admittance control reproduce demonstrated contact behavior better than pose-only, force-observation-only, fixed-force, phase-force, or one-step-force interfaces?

RQ2: How should contact history be represented for continuous polishing, and does a separated GRU-based wrench-history encoder improve force-trajectory tracking relative to unified, current-force, nonrecurrent, and LSTM variants?

RQ3: Do bounded residual policies for sliding velocity and tool-normal alignment improve force tracking, alignment, surface-quality consistency, or task time beyond imitation plus admittance execution?

1.4 Contributions

The contributions are threefold.

1. A VR-calibrated motion–force teaching system transfers synchronized position, orientation, wrench, and image demonstrations from a handheld device equipped with a six-axis force/torque sensor into robot-executable polishing trajectories.
2. A GRU-based force-history encoder represents contact evolution within a force-aware imitation policy that predicts reference pose trajectories and desired force trajectories.
3. A residual RL-augmented admittance execution layer applies bounded sliding-velocity and tool-orientation corrections for precision polishing while retaining the nominal learned motion–force trajectory as the task-level reference.

These contributions are intentionally framed as an integrated precision-polishing system. FAIRE does not claim to introduce force sensing, force prediction, recurrent wrench encoding, admittance control, or residual learning in isolation.

1.5 Scope and Thesis Status

The target task is polishing curved headlight surfaces. Vision is used to provide task-region context; autonomous defect detection or region-of-interest discovery is outside the claimed contribution. The source FAIRE repository is a review-version manuscript draft whose experimental tables, hardware identifiers, network dimensions, controller parameters, and final figures remain incomplete. This thesis preserves that evidentiary boundary. Missing implementation details and measurements are labeled as planned

or TBD, and no numerical advantage is asserted before the corresponding experiment is completed.

1.6 Thesis Organization

Chapter 2 reviews robotic polishing, force-aware imitation, demonstration capture, and compliant residual execution. Chapter 3 defines the FAIRE architecture and problem. Chapter 4 presents VR-calibrated motion–force demonstration transfer. Chapter 5 develops the force-history imitation policy and admittance execution. Chapter 6 formulates the residual precision policies. Chapter 7 specifies the evaluation and ablations. Chapter 8 concludes the thesis and records its limitations.

Chapter 2

Related Work

2.1 Robotic Polishing and Surface Finishing

Robotic surface finishing combines surface representation, path generation, process planning, and contact control. Coverage planners reason about workpiece boundaries, tool footprints, local surface normals, reachability, and collision constraints [10–12]. Force-controlled polishing methods complement geometric planning by regulating contact on curved surfaces [2, 3]. Integrated systems such as RoboGrind demonstrate that inspection, task specification, trajectory generation, and force-controlled execution can form a practical surface-treatment workflow [13].

These methods establish the importance of both geometry and contact. FAIRE addresses a different question: how to reproduce a task-specific motion–force demonstration and then correct its execution reactively. It does not claim that learning eliminates geometric or process knowledge, nor that a demonstrated trajectory guarantees full-surface coverage. Its evaluation is centered on force tracking, contact stability, normal alignment, and surface-quality consistency for specified polishing regions.

2.2 Imitation Learning for Contact-Rich Manipulation

Modern visuomotor imitation policies can generate temporally coherent action sequences from image and robot-state observations [4, 5]. Contact-rich tasks expose an observation gap because the image and pose may not reveal the

force transmitted through the environment. A survey of contact-rich imitation learning describes force, tactile, and compliance information as central to closing this gap [9].

Force-aware methods use interaction measurements in different ways. Admittance Visuomotor Policy Learning predicts a desired force and executes it through an admittance mechanism for general contact-rich manipulation [6]. ForceMimic combines force–motion demonstration capture with learned pose and wrench parameters and hybrid force–position execution [7]. FoAR conditions a high-frequency reactive position policy on force/torque observations rather than using a predicted desired force trajectory as the policy action [8]. MOMA-Force similarly demonstrates visual–force imitation in mobile manipulation [14].

The FAIRE source manuscript additionally compares its intended force-history representation with FAWAM, which predicts future action and wrench trajectories, and identifies G-force as related work on robustness to contact-dynamics variation [15, 16]. These references delimit rather than establish FAIRE’s novelty. FAIRE integrates temporal wrench encoding, desired-force prediction, and compliant execution specifically around polishing precision metrics.

2.3 Motion–Force Demonstration Collection

Contact-rich learning requires synchronized capture of task context, tool motion, and wrench. Kinesthetic teaching obtains demonstrations through the robot, whereas external devices can reduce dependence on a specific robot embodiment. ForceMimic is a direct precedent for a force–motion capture device used in contact-rich imitation [7]. Other surface-treatment interfaces emphasize intuitive task teaching and interactive programming [13].

FAIRE uses a handheld device instrumented with a six-axis force/torque sensor and tracked in a VR coordinate frame. Its method contribution is the calibrated transform chain that maps these measurements into the robot workspace while preserving synchronization with wrench and image streams. The use of six-axis sensing or external tracking alone is not presented as novel.

2.4 Force, Impedance, and Admittance Control

Compliant robot control regulates the relationship between motion and contact force. Impedance control specifies a desired dynamic relation between displacement and force, while admittance control maps measured force into a motion response [17–19]. Admittance is particularly suitable when a learned policy supplies a force reference but the available robot interface accepts Cartesian motion commands.

In FAIRE, admittance control is the explicit bridge between prediction and physical execution. The force-tracking error drives a compliant position correction that is added to the imitated position reference. This formulation avoids describing the learned force as a direct actuator command and keeps safety filtering, saturation, and controller-rate implementation within the low-level execution layer.

2.5 Residual Learning for Precision Correction

Residual learning augments a nominal controller with a bounded learned correction. This decomposition is attractive for polishing because the demonstrated path and force trajectory already define task progress, while local changes in contact or curvature may require small adjustments. Related

Table 2.1: Functional positioning of representative related approaches. Desired force denotes a policy-generated reference rather than force observation alone.

Method	Force observation	Desired force	Compliant execution	Polishing precision focus
Vision-only IL [4, 5]	No	No	No	No
AdmitDiff [6]	Yes	Yes	Yes	No
ForceMimic [7]	Yes	Yes	Yes	No
FoAR [8]	Yes	No	Reactive	No
FAIRE	Yes	Yes	Yes	Yes

compliance-aware and reactive policies use learned fast corrections without assigning the learned residual responsibility for the complete task [20, 21].

FAIRE applies this principle to two variables that directly affect surface processing. A velocity residual scales progress along the demonstrated stroke, and an orientation residual reduces the error between the tool normal and local surface normal. The intended novelty is their integration as polishing-specific precision layers above motion–force imitation and admittance execution, not residual RL as a general algorithm.

2.6 Positioning of FAIRE

The research gap is therefore not the absence of force-aware learning. It is the need for a single precision-oriented polishing framework that connects calibrated human motion–force teaching, temporal force-conditioned imitation, controller-level force-trajectory execution, and bounded velocity and normal-alignment correction, and then evaluates each interface using continuous process metrics.

Chapter 3

FAIRE Overview and Problem Formulation

3.1 System Objective

FAIRE aims to reproduce the continuous contact behavior demonstrated during robotic polishing. Let the robot observation at time t contain an RGB image I_t , a proprioceptive and pose state s_t , and the measured six-axis wrench

$$w_t = [F_x, F_y, F_z, \tau_x, \tau_y, \tau_z]_t. \quad (3.1)$$

The image localizes the task region, the pose state describes the robot configuration and tool motion, and the wrench describes interaction with the workpiece.

The task output is a short horizon of reference position, reference orientation, and desired contact force:

$$\mathcal{A}_t = \{x_{t:t+H}^{\text{ref}}, R_{t:t+H}^{\text{ref}}, F_{t:t+H}^{\text{des}}\}. \quad (3.2)$$

Here H is the prediction horizon. The desired force is a trajectory to be tracked by a compliant controller; it is not a direct joint-torque or actuator-level command.

3.2 Framework Architecture

The framework contains three stages. First, the teaching stage maps handheld motion into the robot base frame through an explicit calibration chain and stores synchronized image, pose, and wrench measurements. Second,

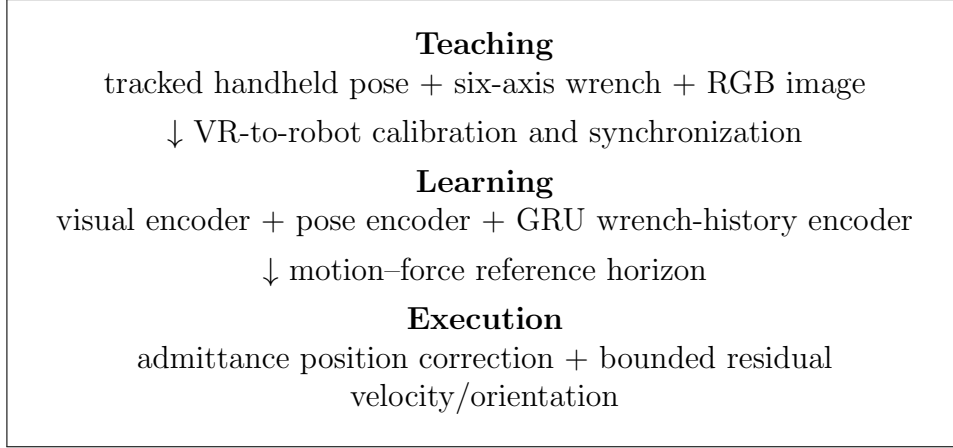


Figure 3.1: FAIRE architecture. A final implementation figure remains planned in the source manuscript.

the learning stage encodes the current visual and pose context together with a wrench-history window. Third, the execution stage uses admittance control to realize the desired force and residual policies to correct velocity and orientation.

3.3 Temporal Contact Representation

Polishing contact is history dependent. The same instantaneous normal force can occur during initial contact, stable sliding, or recovery from excessive load. FAIRE therefore uses the wrench window

$$w_{t-K:t} = \{w_{t-K}, \dots, w_t\}, \quad (3.3)$$

where $K + 1$ is the history length. The recurrent representation is

$$z_t^F = \text{GRU}(w_{t-K:t}). \quad (3.4)$$

Separate visual and pose encoders produce $z_t^I = f_I(I_t)$ and $z_t^P = f_P(s_t)$. Their fused latent state,

$$z_t = \text{Fusion}(z_t^I, z_t^P, z_t^F), \quad (3.5)$$

conditions the motion-force policy. Separation is important to the planned ablation: unified and split encoders, current and historical force inputs, and

recurrent alternatives can be compared while holding the action and controller interfaces fixed.

3.4 Nominal and Residual Responsibilities

The nominal policy π_θ produces the task-level reference,

$$\pi_\theta(z_t) \mapsto \mathcal{A}_t. \quad (3.6)$$

The admittance controller maps force error into the compliant position adjustment needed to realize this reference. Residual policies then make bounded precision corrections:

$$v_t^{\text{cmd}} = \alpha_t v_t^{\text{ref}}, \quad (3.7)$$

$$R_t^{\text{cmd}} = R_t^{\text{ref}} \Delta R_t^{\text{RL}}. \quad (3.8)$$

The velocity scale α_t and residual rotation ΔR_t^{RL} are restricted by safety and smoothness limits. They do not generate a new path, target region, or force trajectory.

3.5 Design Principles

Four principles constrain the framework.

1. **Synchronized multimodal teaching:** pose, orientation, image, and wrench must share a common time base and coordinate convention.
2. **Explicit execution semantics:** the predicted desired force must be realized through a specified controller rather than described ambiguously as a robot action.
3. **Temporal contact context:** the force representation must distinguish contact phases that are not identifiable from a single sample.

4. **Bounded correction:** RL is limited to precision adjustments; nominal task execution remains interpretable through the demonstrated motion–force reference.

3.6 Assumptions and Safety Boundary

The target polishing region is specified before demonstration and evaluation. The framework assumes that the task remains within the demonstrated visual, geometric, and contact distribution and that the robot provides a Cartesian command interface compatible with admittance correction. Wrench bias compensation, frame transformation, filtering, action limits, force limits, velocity limits, workspace limits, and emergency-stop logic are required implementation elements. Their verified numerical settings are [\[Planned/TBD: TBD in the FAIRE source repository\]](#).

Chapter 4

VR-Calibrated Motion–Force Demonstration Collection

4.1 Teaching-System Purpose

The demonstration system must capture how a human moves the polishing tool and how the tool interacts with the surface. FAIRE uses a handheld teaching device instrumented with a six-axis force/torque sensor. A VR tracker measures the device pose, and an RGB camera records visual task context. These streams are synchronized so that each visual and kinematic state is paired with the wrench produced at the same stage of contact.

The device is intentionally separated from the robot. This arrangement enables natural task demonstration in the workpiece area, but it creates a calibration problem: tracker poses are measured in a VR tracking frame, whereas the learned trajectory must be executed in the robot base frame.

4.2 Coordinate Frames and Pose Transfer

Let B , V , T , and E denote the robot base, VR tracking, tracker, and end-effector/tool frames. The demonstrated tool pose expressed in the robot base frame is

$${}^B T_E^{\text{demo}}(t) = {}^B T_V {}^V T_T(t) {}^T T_E. \quad (4.1)$$

The time-varying transform ${}^V T_T(t)$ is measured by the tracking system. The constant transform ${}^T T_E$ describes the mechanical offset from tracker to tool, and ${}^B T_V$ aligns the VR tracking coordinates with the robot base.

Robot base $B \xleftarrow{{}^B T_V}$ VR frame $V \xleftarrow{{}^V T_T(t)}$ tracker $T \xleftarrow{{}^T T_E}$ tool E
 synchronized RGB image I_t , tool pose (x_t, R_t) , and wrench w_t

Figure 4.1: Transform chain for VR-to-robot demonstration transfer. Numerical calibration results and the final apparatus drawing remain planned.

Equation (4.1) makes the transfer convention explicit. It also exposes the calibration components that must be validated separately. Errors in ${}^B T_V$ shift or rotate the entire demonstration in the robot workspace, whereas errors in ${}^T T_E$ produce a tool-dependent offset. The final implementation should report the calibration observations, estimator, residual translational and rotational errors, and validation procedure. These details are [Planned/TBD: not specified in the source FAIRE manuscript].

4.3 Wrench Processing

The force/torque sensor measures

$$w_t = [F_x, F_y, F_z, \tau_x, \tau_y, \tau_z]_t. \quad (4.2)$$

Raw readings require bias compensation before they are used for learning or control. The compensated wrench must then be represented in a consistent frame. If the sensor-frame wrench is transformed to the tool or robot frame, both force and torque must follow the appropriate spatial wrench transformation; simply rotating the force vector is insufficient when the reference point changes.

The FAIRE manuscript states that bias compensation and frame normalization are performed but does not define the calibration pose, averaging

interval, gravity/tool-load compensation, reference point, or filter. The thesis therefore treats these as required implementation records rather than assigning unverified values. The same convention must be used for demonstration targets, policy observations, measured execution forces, and evaluation metrics.

4.4 Synchronization and Dataset Construction

After calibration and preprocessing, a demonstration episode is stored as

$$D = \{I_t, x_t, R_t, w_t\}_{t=1}^T, \quad (4.3)$$

where x_t and R_t are the tool position and orientation expressed in the robot workspace. A usable dataset also requires a timestamp for each stream, episode boundaries, and enough metadata to reproduce the coordinate convention and preprocessing.

The final synchronization method is [\[Planned/TBD: TBD\]](#). It should document whether measurements are hardware-triggered, timestamp-aligned, or resampled; the image, tracker, wrench, and controller rates; interpolation rules; the maximum accepted skew; and the treatment of missing samples. Temporal alignment is especially important because the learning target contains both motion and force. A systematic delay can make the policy associate a pose with a force that was generated earlier or later in the stroke.

4.5 Segmentation and Training Windows

Continuous episodes are divided into history-and-target samples. At time t , the input includes current visual and pose information and the wrench history $w_{t-K:t}$. The target contains a future horizon of reference positions, orientations, and desired forces from t through $t + H$. Samples near episode boundaries

require either shortened windows, padding, or exclusion; the selected rule must be shared by all ablation variants.

Train, validation, and test partitions should be defined at the episode or workpiece-trial level rather than by randomly mixing adjacent windows. Otherwise, nearly identical temporal neighbors can leak across partitions. The split, number of demonstrations, window stride, K , H , normalization statistics, and augmentation policy remain [Planned/TBD: TBD pending the final dataset].

4.6 Demonstration Quality and Safety

The collection protocol should reject or label demonstrations with sensor saturation, tracking loss, unsafe load, accidental contact, missing images, or timestamp discontinuities. The robot must not replay a transferred demonstration until the transformed trajectory has passed workspace, reachability, collision, velocity, and force-limit checks. Calibration enables coordinate transfer; it does not by itself establish safe executability.

For reproducibility, each accepted episode should include the workpiece identifier, target region, sensor zeroing state, calibration version, tool configuration, start pose, sampling metadata, and operator annotation. These fields are listed in Appendix A.

Chapter 5

Force-History Imitation and Admittance Execution

5.1 Force-Aware Observation

The imitation policy receives complementary visual, kinematic, and interaction observations. The image I_t describes the specified task region, the robot state s_t describes the current tool pose and proprioception, and the wrench history $w_{t-K:t}$ describes recent contact evolution. Vision is not used as an autonomous defect detector in this thesis.

Separate encoders compute

$$z_t^I = f_I(I_t), \quad (5.1)$$

$$z_t^P = f_P(s_t), \quad (5.2)$$

$$z_t^F = \text{GRU}(w_{t-K:t}). \quad (5.3)$$

The force branch uses the full six-axis wrench rather than normal force alone so that tangential force and torque can provide evidence about sliding, tool tilt, or partial contact. The exact image backbone, state variables, layer dimensions, and GRU hidden size are [Planned/TBD: TBD].

5.2 Motion–Force Trajectory Prediction

The fused representation is mapped to a short reference horizon:

$$\pi_\theta(z_t) \rightarrow \{x_{t:t+H}^{\text{ref}}, R_{t:t+H}^{\text{ref}}, F_{t:t+H}^{\text{des}}\}. \quad (5.4)$$

Position and orientation define the demonstrated motion, while F^{des} defines the demonstrated contact-force behavior to be realized by the controller. In the

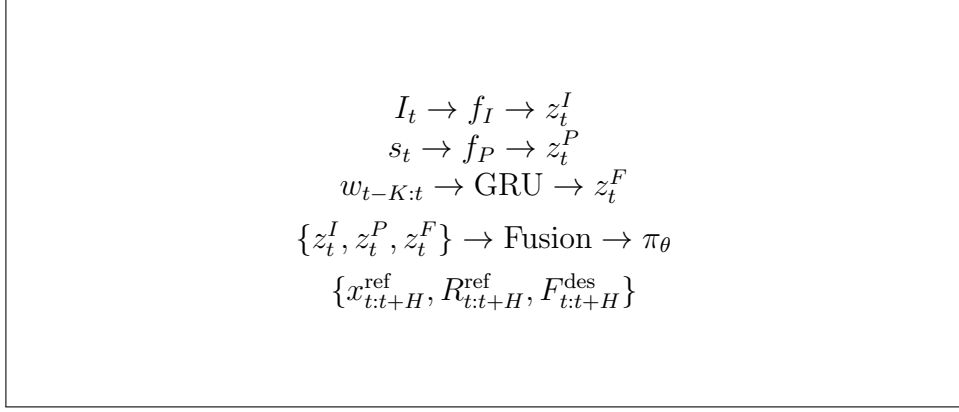


Figure 5.1: GRU-based force-history encoder and motion-force policy. Final network dimensions and the implementation-specific backbone remain TBD.

experimental notation of Chapter 7, the six-dimensional motion representation and three-dimensional force form a 9D action at each horizon step.

Predicting a horizon rather than a single next sample provides a reference segment that can be executed smoothly and supports the planned comparison between one-step force prediction and force-trajectory prediction. The source manuscript does not specify how overlapping horizons are combined online. Action selection, temporal ensembling, replanning frequency, and horizon interruption are therefore [Planned/TBD: TBD].

5.3 Supervised Learning Objective

Training samples are drawn from the calibrated demonstration dataset in Eq. (4.3). The multitask objective is

$$\mathcal{L}_{\text{IL}} = \lambda_x \mathcal{L}_x + \lambda_R \mathcal{L}_R + \lambda_F \mathcal{L}_F, \quad (5.5)$$

where $\mathcal{L}_x$, $\mathcal{L}_R$, and $\mathcal{L}_F$ penalize position, orientation, and force-trajectory prediction error. Position and force can be trained with component-wise or norm-based regression losses. Orientation requires a representation-aware error, such as a geodesic rotation loss, if the output is a rotation matrix or quaternion.

The source manuscript deliberately leaves the exact loss definitions and weights open. The final thesis must report output normalization, orientation parameterization, force channels, masking of padded targets, optimizer, learning-rate schedule, batch size, epoch or step count, seed count, checkpoint selection, and regularization. Until those records are available, Eq. (5.5) is a framework-level specification rather than a complete reproduction recipe.

5.4 Why Force History Is Separated

A unified encoder can learn correlations among pose and force, but it does not explicitly preserve the distinct temporal structure of contact. FAIRE separates the force branch so that its representation can be ablated independently. The experimental comparison includes a unified current-sample MLP, split current-sample MLP, split history MLP, split history LSTM, and split history GRU. Each variant retains the same 9D trajectory output and admittance execution, isolating the effect of force representation.

The claim tested by this comparison is that temporal force conditioning improves contact reproduction. The GRU architecture itself is not claimed as a new recurrent model.

5.5 Admittance-Based Force-Trajectory Execution

The low-level controller converts desired force into a Cartesian correction according to

$$M_d \Delta \ddot{x}_t + D_d \Delta \dot{x}_t + K_d \Delta x_t = F_t^{\text{meas}} - F_t^{\text{des}}, \quad (5.6)$$

where M_d , D_d , and K_d are virtual inertia, damping, and stiffness matrices in the controlled contact frame. The corrected position command is

$$x_t^{\text{cmd}} = x_t^{\text{ref}} + \Delta x_t. \quad (5.7)$$

This equation defines the semantics of the learned force output. The policy predicts a reference force; the controller observes the tracking error and moves compliantly to reduce it. The imitation policy is not itself an admittance controller, and the force value is not directly mapped to actuator torque.

5.6 Discrete Execution and Safety Requirements

Implementation requires discretizing Eq. (5.6) at the controller rate, filtering the measured wrench, selecting controlled axes, and limiting Δx_t , $\Delta \dot{x}_t$, and total commanded motion. Contact establishment and release also require explicit state handling so that the controller does not integrate noise while the tool is away from the surface.

The final controller specification must include M_d , D_d , K_d , update rate, filter, force frame, saturation values, contact threshold, anti-windup or state-reset rule, and robot command interface. These quantities are [Planned/TBD: TBD]; no claim of robustness to varying or learned admittance mass and damping is made.

5.7 Online Execution Sequence

At each policy update, FAIRE performs the following conceptual sequence:

1. acquire and preprocess the current image, robot state, and wrench;
2. update the wrench-history buffer and compute z_t^F ;
3. predict a motion–force reference horizon;
4. select the current reference sample and compute force error;
5. integrate the admittance dynamics subject to safety limits;

6. apply the residual velocity and orientation corrections defined in Chapter 6;
7. transmit the resulting Cartesian command through the robot's safe command interface.

Policy and controller updates may operate at different rates. Their exact scheduling is part of the pending implementation record.

Chapter 6

Residual Reinforcement Learning for Precision Correction

6.1 Role of the Residual Layer

The nominal motion–force policy and admittance controller define a complete polishing execution path. Residual RL is added only to correct variables that strongly affect polishing precision under local contact and curvature variation. This decomposition narrows the learning problem and keeps task progress anchored to the demonstration.

The residual layer contains two policies. The velocity policy adjusts progress along the reference stroke, and the orientation policy adjusts alignment between the tool and surface normals. Their actions are bounded and regularized for smoothness. Neither policy selects the target region, generates the nominal path, nor replaces the desired force trajectory.

6.2 Adaptive Sliding-Velocity Policy

Let

$$e_F(t) = F_t^{\text{meas}} - F_t^{\text{des}} \quad (6.1)$$

denote the force-tracking error. The velocity policy observes this error, recent wrench history, and robot state:

$$\alpha_t = \pi_v(e_F(t), w_{t-K:t}, s_t). \quad (6.2)$$

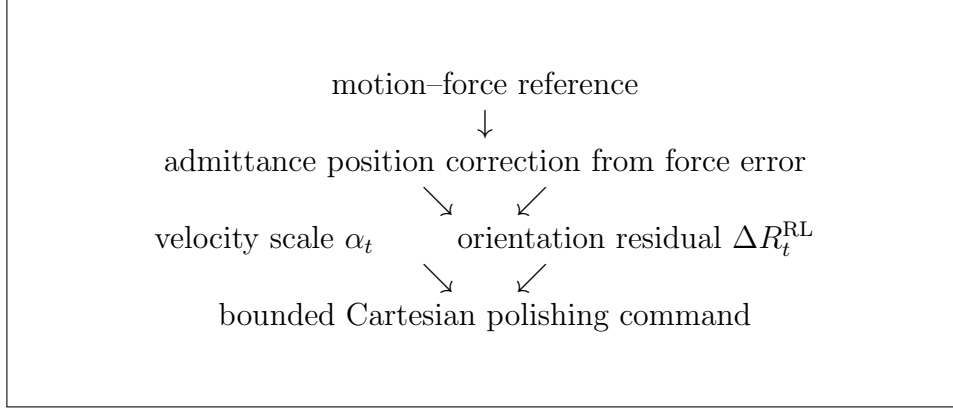


Figure 6.1: Residual RL-augmented execution. The residual policies correct precision variables while nominal execution remains with imitation and admittance control.

The commanded tangential velocity is

$$v_t^{\text{cmd}} = \alpha_t v_t^{\text{ref}}, \quad (6.3)$$

where α_t is constrained to a safe interval. Reducing progress can give the admittance loop time to recover from a force deviation, while increasing progress can reduce excessive dwell when contact is stable. Because speed also affects material-removal rate, the policy objective must balance force tracking, finish consistency, execution time, smoothness, and safety.

A generic planned reward decomposition is

$$r_t^v = -c_F(e_F) - c_{\Delta\alpha}(\alpha_t - \alpha_{t-1}) - c_{\text{safety}} + r_{\text{progress}}, \quad (6.4)$$

where each term must be defined and scaled from measured quantities. Equation (6.4) records the intended objective categories, not finalized reward weights.

6.3 Tool-Normal Alignment Policy

Let n_t^{tcp} be the tool-center-point normal and n_t^{surf} the estimated local surface normal. Their angular discrepancy is

$$e_n(t) = \cos^{-1}(\text{clip}((n_t^{\text{tcp}})^\top n_t^{\text{surf}}, -1, 1)). \quad (6.5)$$

The orientation policy produces a bounded residual rotation:

$$\Delta R_t^{\text{RL}} = \pi_R(n_t^{\text{tcp}}, n_t^{\text{surf}}, w_{t-K:t}, s_t), \quad (6.6)$$

$$R_t^{\text{cmd}} = R_t^{\text{ref}} \Delta R_t^{\text{RL}}. \quad (6.7)$$

The multiplication order in Eq. (6.7) defines the residual in the local reference frame. A different implemented convention must be documented explicitly.

A planned orientation reward can combine normal-alignment error, force tracking, rotation magnitude, temporal smoothness, and safety:

$$r_t^R = -c_n(e_n) - c_F(e_F) - c_R(\Delta R_t^{\text{RL}}) - c_{\Delta R} - c_{\text{safety}}. \quad (6.8)$$

The final estimator for n_t^{surf} , reward terms, and weights remain [Planned/TBD: TBD].

6.4 Training Interface

The source FAIRE manuscript does not name the RL algorithms or state whether the policies are trained in simulation, from logged data, on the physical robot, or through a staged combination. Consequently, this thesis does not infer an algorithm. The final report must specify the Markov state, action parameterization, observation normalization, algorithm, network size, discount, optimizer, replay or rollout scheme, environment randomization, termination conditions, and number of training steps.

Safe training should begin with a verified nominal controller and conservative action bounds. Physical-robot exploration, if used, requires force, velocity, orientation, workspace, and cumulative-load limits plus an independent stop mechanism. A residual action must be clipped before command composition, not only discouraged by reward.

6.5 Policy Interaction

The two residual policies may interact because velocity affects force tracking and orientation affects contact distribution. The planned ablation therefore evaluates no residual policy, velocity only, orientation only, and both policies. All four variants retain the same imitation representation, force-trajectory output, and admittance controller so that performance changes can be attributed to the correction layer.

Joint execution also requires a specified command order. The conceptual FAIRE order computes the admittance position correction from force error, scales tangential reference velocity, applies the bounded orientation residual, and then enforces final safety limits. Controller implementation, update rates, and arbitration rules remain [Planned/TBD: TBD].

6.6 Expected Failure Modes

Potential failures include velocity oscillation, residual saturation, inaccurate surface normals, overcorrection on high curvature, conflict between force and orientation objectives, and distribution shift outside the demonstrations or RL training environment. Evaluation should report safety interventions and saturation frequency alongside average performance. A gain in mean polishing quality is insufficient if it is accompanied by rare excessive-force events.

Chapter 7

Experimental Evaluation

7.1 Evaluation Strategy

All planned experiments use curved headlight surfaces. The target polishing region is specified before demonstration and evaluation; autonomous region or defect detection is not treated as a contribution. The evaluation directly follows the three research questions in Chapter 1:

1. compare force-action interfaces under a fixed force-history representation;
2. compare motion–force representations under a fixed trajectory action and admittance controller;
3. compare residual precision policies under a fixed FAIRE imitation and execution stack.

This staged design changes one interface at a time. It avoids attributing a performance difference to the GRU when the compared method also uses a different action, or attributing a residual-policy effect to a changed admittance controller.

7.2 Experimental Setup

The robot carries a polishing tool and a six-axis force/torque sensor, and an RGB camera supplies visual context. The handheld teaching device described in Chapter 4 records synchronized pose, wrench, and image demonstrations. Each robot trial begins from a standardized pre-contact pose and ends at the

policy horizon, task-completion condition, or safety limit. Surface appearance is recorded under controlled conditions before and after polishing.

The source FAIRE manuscript does not identify the robot, tool, force/torque sensor, camera, abrasive, workpiece set, sampling rates, control rates, number of demonstrations, or number of evaluation trials. It also does not state the target-region protocol, termination criteria, or safety thresholds. These fields are [Planned/TBD: TBD and must be filled from the final implementation and experiment logs].

7.3 Experimental Notation

The six-dimensional motion state, three-dimensional force, and six-axis wrench are

$$q_t = [x_t, y_t, z_t, \omega_{x,t}, \omega_{y,t}, \omega_{z,t}], \quad (7.1)$$

$$f_t = [F_{x,t}, F_{y,t}, F_{z,t}], \quad (7.2)$$

$$w_t = [F_{x,t}, F_{y,t}, F_{z,t}, \tau_{x,t}, \tau_{y,t}, \tau_{z,t}]. \quad (7.3)$$

The proposed action is a horizon of 9D motion–force references,

$$a_t = [q_{t:t+H}^{\text{ref}}, f_{t:t+H}^{\text{des}}], \quad (7.4)$$

and the force-history observation is $w_{t-K:t}$. Unless an ablation states otherwise, a force-aware variant uses a split motion–force representation and the GRU-based six-axis wrench-history encoder.

7.4 Metrics

7.4.1 Force-Trajectory Tracking

Normal-force root-mean-square error is evaluated after aligning an executed trial with its demonstration-derived reference by normalized task progress:

$$F_n\text{-RMSE} = \sqrt{\frac{1}{N} \sum_{i=1}^N (F_{n,i}^{\text{meas}} - F_{n,i}^{\text{des}})^2}. \quad (7.5)$$

Force correlation (F-Corr) measures temporal similarity between the aligned measured and desired normal-force profiles. Overshoot is the maximum positive value of $F_n^{\text{meas}} - F_n^{\text{des}}$. Contact Loss is the fraction of evaluated samples below a predefined normal-force contact threshold.

7.4.2 Alignment and Process Quality

Normal-alignment error (NAE) is the angular difference in Eq. (6.5). Polishing-quality improvement (PQI) compares controlled pre- and post-polishing measurements. The FAIRE manuscript leaves PQI measurement-agnostic so that it can be instantiated with verified gloss, haze, roughness, or image-based surface-quality measurements. Consistency measures the spatial uniformity of the post-polishing finish, and Task Time is elapsed execution time.

The final evaluation must specify the contact threshold, progress-alignment algorithm, surface-normal estimator, PQI instrument or image protocol, spatial sampling pattern, consistency definition, units, aggregation statistic, confidence interval, and treatment of failed trials. These items are [Planned/TBD: TBD].

7.5 RQ1: Force-Action Interface and Admittance Execution

The first ablation asks how the demonstrated force trajectory should be represented by the policy and used during execution. Pose-only Policy excludes force observations and force output. Force-Observation Pose Policy adds wrench history but still predicts pose only. Fixed-Force Admittance uses a constant target F_0 . Phase-Force Admittance uses a demonstration-derived phase profile $\bar{f}^{\text{demo}}(s)$. One-Step Force Action predicts only the current desired force, whereas FAIRE predicts a force trajectory aligned with the motion horizon.

All force-aware rows use the same GRU-based six-axis wrench-history encoder; residual RL is disabled. Thus, Table 7.1 isolates the action and low-level execution interfaces.

Table 7.1: Planned ablation of force-action interface and admittance execution. Dashes indicate measurements not yet reported in the FAIRE source manuscript.

Method	Policy output	Force target	Execution	F_n -RMSE $\downarrow$	F-Corr $\uparrow$	Overshoot $\downarrow$	Contact Loss $\downarrow$	PQI $\uparrow$
Pose-only Policy	$q_{t:t+H}^{\text{ref}}$	None	Position	—	—	—	—	—
Force-Obs. Pose Policy	$q_{t:t+H}^{\text{ref}}$	None	Position	—	—	—	—	—
Fixed-Force Admittance	$q_{t:t+H}^{\text{ref}}$	Constant F_0	Admittance	—	—	—	—	—
Phase-Force Admittance	$q_{t:t+H}^{\text{ref}}$	$\bar{f}^{\text{demo}}(s)$	Admittance	—	—	—	—	—
One-Step Force Action	$q_{t:t+H}^{\text{ref}}, f_t^{\text{des}}$	Predicted step	Admittance	—	—	—	—	—
FAIRE 9D Trajectory	$q_{t:t+H}^{\text{ref}}, f_{t:t+H}^{\text{des}}$	Predicted trajectory	Admittance	—	—	—	—	—

7.6 RQ2: Motion–Force Representation

The second ablation asks how force history should be encoded when every method predicts the same 9D trajectory action and uses the same admittance controller. Unified Current MLP concatenates the current motion and force sample. Split Current MLP uses separate current-sample branches. Split History MLP flattens or otherwise encodes a window without recurrence. LSTM and GRU variants use recurrent force-history encoders. The motion encoder, fusion network, policy backbone, training split, action horizon, and controller must remain fixed.

Table 7.2: Planned ablation of motion-force representation.

Method	Input	Motion encoder	Force observation	Force encoder	F_n -RMSE ↓	F-Corr ↑	Overshoot ↓	Contact Loss ↓
Unified	$[q_t, f_t]$	shared MLP	current	shared MLP	—	—	—	—
Split Current MLP	q_t, f_t	MLP	current	MLP	—	—	—	—
Split History MLP	$q_t, f_{t-K:t}$	MLP	history	MLP	—	—	—	—
Split History LSTM	$q_t, f_{t-K:t}$	MLP	history	LSTM	—	—	—	—
FAIRE History GRU	$q_t, f_{t-K:t}$	MLP	history	GRU	—	—	—	—

The hypothesis is that force-history conditioning improves reproduction of demonstrated contact behavior during continuous polishing. This is an experimental hypothesis, not a result reported by the current draft.

7.7 RQ3: Residual Precision Policies

The third ablation measures the contribution of velocity and orientation correction. Every row uses the FAIRE History GRU representation, 9D trajectory action, policy backbone, and admittance controller. Only the two residual policies vary.

Table 7.3: Planned ablation of residual RL precision policies.

Method	Velocity	Orientation	F_n -RMSE ↓	NAE ↓	PQI ↑	Consistency ↑	Task Time ↓
FAIRE without residual RL	No	No	—	—	—	—	—
Velocity policy only	Yes	No	—	—	—	—	—
Orientation policy only	No	Yes	—	—	—	—	—
FAIRE full	Yes	Yes	—	—	—	—	—

7.8 Protocol and Statistical Reporting

Trial ordering should be randomized or counterbalanced across methods to reduce confounding from abrasive wear, surface condition, and operator setup. Methods should use identical target regions where repeated processing is physically valid; otherwise, comparable regions and a documented assignment scheme are required. Demonstrations, hyperparameter selection, and final test trials must be separated.

For each metric, the thesis should report the number of independent workpieces or regions, trials per condition, central tendency, dispersion, confidence intervals, and the planned statistical test. Paired analysis is preferable when methods can be evaluated on matched regions. Failed or safety-stopped trials must remain visible through success and intervention counts rather than being silently removed from continuous metrics.

7.9 Qualitative Analysis

Planned qualitative evidence includes representative robot trajectories, desired-versus-measured normal-force traces, examples of TCP and surface-normal alignment, and controlled before/after surface observations. Visual examples complement but do not replace force, alignment, and surface-quality measurements. VR-to-robot calibration is used consistently by all methods and is documented as a pipeline component rather than presented as a separate performance baseline.

7.10 Current Result Status

All quantitative cells remain blank because the FAIRE repository contains no completed measurements or statistical analysis. Accordingly, this thesis does not state that one method outperforms another. The tables constitute

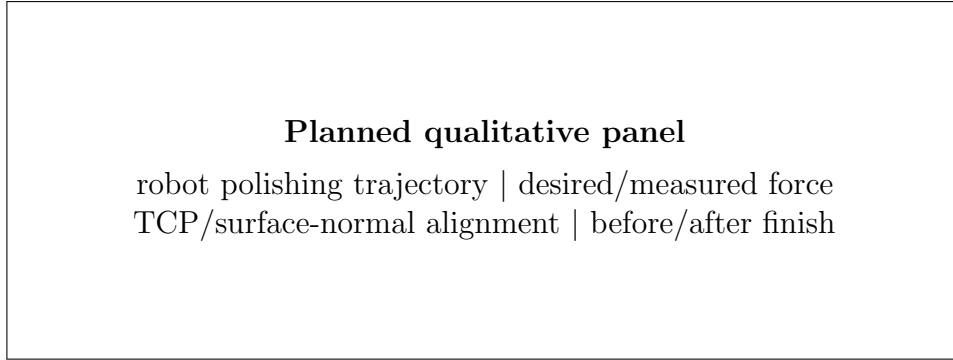


Figure 7.1: Planned qualitative results on curved headlight surfaces. No result image is available in the source repository.

a preregistered comparison structure to be populated only from traceable experiment logs and inspection data.

Chapter 8

Conclusions and Future Work

8.1 Conclusions

This thesis presented FAIRE, a precision-oriented framework for force-aware imitation with reactive execution in robotic polishing. Its first component is a VR-calibrated handheld teaching system that transfers synchronized position, orientation, wrench, and image demonstrations into the robot workspace. Its second component is a force-aware imitation policy whose GRU encoder represents recent six-axis wrench history and whose output contains both motion references and a desired force trajectory. Its third component is an execution stack in which admittance control realizes the learned force reference through compliant Cartesian correction and bounded residual policies adjust sliding velocity and tool-normal alignment.

The framework makes a deliberate separation of responsibilities. The imitation policy defines task-level motion and contact references. The admittance controller converts force error into compliant motion. Residual RL supplies precision corrections without replacing the task. This separation makes the learned force output physically interpretable and enables controlled ablations of observation, action, representation, controller, and residual policy.

The planned experiments evaluate force tracking, contact stability, normal alignment, polishing-quality improvement, spatial consistency, and task time on curved headlight surfaces. The source FAIRE manuscript does not yet provide numerical results. Therefore, no quantitative superiority claim is made in this version of the thesis.

8.2 Limitations

The principal limitation is evidentiary completeness. Hardware identifiers, calibration accuracy, synchronization details, network dimensions, learning hyperparameters, admittance parameters, residual RL algorithms, safety bounds, trial counts, and quantitative results remain TBD. The current manuscript defines the intended system and experimental interfaces but is not yet sufficient for independent replication.

The framework also assumes a specified target polishing region and does not solve autonomous defect detection. Its demonstrations and policies may not generalize to unseen geometries, materials, tools, abrasives, or contact regimes. A predicted desired force does not guarantee safe physical execution; performance depends on sensor quality, frame consistency, controller tuning, saturation, and independent safety logic. Surface-normal estimation error can degrade orientation correction, and the velocity and orientation residuals may interact.

8.3 Future Work

Immediate work should complete the implementation record and populate the preregistered ablations with repeated real-robot trials. The calibration transform should be validated quantitatively, multimodal delay should be measured, and the discrete admittance implementation should be tested under conservative force limits before learned residuals are enabled.

The evaluation should then expand across multiple curved surfaces, target regions, tools, abrasive conditions, and initial-contact offsets. Surface quality should be measured with a repeatable instrument or controlled vision protocol and related to the tracked force, velocity, and alignment histories. Reporting interventions, saturation, and worst-case force is as important as reporting

average quality.

Longer-term extensions include uncertainty-aware force prediction, safety-filtered residual actions, online contact-frame estimation, and adaptation across workpiece geometry and tool wear. Autonomous task-region localization may be integrated as a separate upstream module, but it should remain distinguishable from FAIRE’s core contribution: faithful, compliant reproduction and precision correction of demonstrated polishing contact behavior.

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Appendix A

Reproducibility and Completion Checklist

A.1 Demonstration Record

Each episode should contain:

- timestamps for image, tracker, wrench, robot, and command streams;
- raw and calibrated tool pose with explicit frame names and transform version;
- raw, bias-compensated, filtered, and transformed six-axis wrench;
- RGB observations, target-region metadata, and workpiece identifier;
- episode boundaries, contact phase, safety events, and acceptance or rejection status;
- tool, abrasive, sensor-zeroing, and operator metadata.

A.2 Implementation Parameters to Report

- VR-to-robot and tracker-to-tool calibration procedures and residual errors;
- stream rates, synchronization method, interpolation, filtering, and maximum skew;
- image and pose encoders, GRU/LSTM dimensions, history length K , horizon H , fusion and prediction heads;
- loss definitions, weights, normalization, optimizer, schedule, split, seed count, and checkpoint rule;

- admittance axes, frame, discrete update, M_d , D_d , K_d , contact detection, saturation, and reset behavior;
- RL algorithms, observations, actions, rewards, bounds, rates, training environment, randomization, and stopping rule.

A.3 Experiment Fields to Complete

- robot, polishing tool, force/torque sensor, camera, curved headlight set, and abrasive consumables;
- demonstration count, trial count, target-region assignment, randomized order, and termination conditions;
- contact threshold, normal estimator, PQI protocol, consistency metric, timing definition, and statistical analysis;
- all cells in Tables 7.1, 7.2, and 7.3;
- representative trajectories, force traces, alignment examples, before/after observations, and failure cases.

A.4 Evidence Status

The present thesis is an expanded, template-adapted version of the FAIRE review draft. Every numerical result and implementation-specific value absent from that repository remains planned or TBD. Before submission, each placeholder should be replaced by a value traceable to code, configuration, calibration output, experiment log, or inspection record.

국문요약

FAIRE: 로봇 폴리싱을 위한 힘 인지 모방학습 및 반응형 실행

로봇 폴리싱은 목표 영역에 도달하거나 기하 경로를 완주하는 것만으로 충분하지 않다. 균일한 표면 마감 품질을 얻기 위해서는 작업 중의 힘 궤적, 미끄럼 속도, 그리고 폴리싱 공구와 작업물 표면 사이의 정렬을 연속적으로 재현해야 한다. 비전 기반 모방학습 정책은 시연된 움직임을 재현할 수 있지만 접촉 상태를 직접 관측하지 못한다. 반면 힘 인지 정책은 접촉 여부를 구분하는데 그치지 않고, 사람의 시연에서 얻은 상호작용을 로봇이 안전하고 유연하게 실행할 수 있는 기준 신호로 변환해야 한다.

본 논문은 정밀 로봇 폴리싱을 위한 힘 인지 모방학습 및 반응형 실행 프레임워크 FAIRE를 제안한다. 먼저 6축 힘/토크 센서가 장착되고 가상현실 좌표계에서 추적되는 휴대형 교시 장치를 사용하여 위치, 자세, 렌치 및 영상 데이터를 동기화하여 수집한다. 명시적인 VR-로봇 보정을 통해 이 시연을 로봇 기준 좌표계로 변환한다. 이후 gated recurrent unit은 일정 구간의 렌치 이력을 부호화하여 접촉 상태의 시간적 변화를 표현한다. 영상, 로봇 상태 및 힘 이력 특징을 결합한 정책은 단기 구간의 위치·자세 기준 궤적과 목표 힘 궤적을 함께 예측한다.

학습된 목표 힘은 액추에이터에 직접 전달되는 명령이 아니라 제어기 수준의 기준 신호이다. 실행 단계에서 어드미턴스 제어기는 측정 힘과 목표 힘의 오차를 유연한 직교좌표 위치 보정으로 변환한다. 또한 두 개의 제한된 잔차 강화학습 정책이 정밀 보정 계층으로 동작한다. 첫 번째 정책은 접촉 상태에 따라 미끄럼 속도를 조절하고, 두 번째 정책은 공구 중심점 법선과 추정 표면 법선 사이의 정렬 오차를 줄이도록 공구 자세를 보정한다. 작업 수준의 실행 책임은 모방학습 정책과 어드미턴스 제어기에 유지된다.

평가는 세 가지 연구 질문을 중심으로 구성한다. 시간에 따라 변하는 운동-힘 행동이 접촉 재현성을 향상시키는지, 어떠한 힘 이력 표현이 시연된 힘 궤적을 가장 잘 추종하는지, 그리고 잔차 속도·자세 보정이 폴리싱 정밀도를 추가로 향상시키는지 검증한다. 곡면 헤드라이트를 대상으로 법선 힘 오차,

힘 궤적 상관도, 오버슈트, 접촉 손실, 법선 정렬 오차, 표면 품질 향상, 공간적 균일도 및 작업 시간을 측정하도록 실험을 설계하였다. 기존 FAIRE 원고에는 아직 완료된 실험 수치가 없으므로, 본 논문은 정량적 결론과 확인되지 않은 구현 변수를 추정하지 않고 계획 또는 TBD로 명시한다.

주요어: 로봇 폴리싱, 모방학습, 힘/토크 센싱, 힘 제어, 어드미턴스 제어, 잔차 강화학습